Variance bounds for proportions and sums

Abstract

We record bounds on the variance $p(1-p)/n$ of a sample proportion, a concavity property of its square root, a uniform bound in n, and a bound for the sample variance of a sum of two data vectors.

1 Setting

Throughout, $n \geq 1$ is an integer and I denotes the unit interval.

Definition 1.1. For $p \in \mathbb{R}$, let $V_n(p) = p(1-p)/n$; for $p \in I$ this is the variance of the proportion of successes in n independent trials with success probability p.

Definition 1.2. For $m \geq 2$ and $u \in \mathbb{R}^m$, let $\bar{u} = \frac{1}{m}\sum_{i=1}^{m} u_i$ and $s^2(u) = \frac{1}{m-1}\sum_{i=1}^{m}(u_i - \bar{u})^2$.

2 Results

Lemma 2.1. *For every $p \in \mathbb{R}$, $p(1-p) = \frac{1}{4} - (p - \frac{1}{2})^2$. In particular $V_n(p) \leq \frac{1}{4n}$.*

Proof. Expanding, $\frac{1}{4} - (p-\frac{1}{2})^2 = \frac{1}{4} - p^2 + p - \frac{1}{4} = p(1-p)$. Since $(p-\frac{1}{2})^2 \geq 0$ and $n > 0$, the bound follows. □

Lemma 2.2. *For every $p \in I$, $V_n(p) > 0$.*

Proof. For $p \in I$ both p and $1-p$ are positive, and $n > 0$. □

Proposition 2.3. *For all $p, q \in I$, $\sqrt{V_n\left(\frac{p+q}{2}\right)} \geq \frac{1}{2}\left(\sqrt{V_n(p)} + \sqrt{V_n(q)}\right)$.*

Proof. It is well known that the function $p \mapsto \sqrt{p(1-p)}$ is concave on I. The claim is the inequality defining concavity at the midpoint of p and q, divided by $\sqrt{n}$. □

Lemma 2.4. *For every $p \in I$, $V_n(p) \leq \frac{p}{4n}$.*

Proof. By Lemma 2.1, $V_n(p) = \frac{p(1-p)}{n} \leq \frac{p}{4n}$. □

Proposition 2.5. *There is a constant C such that $V_n(p) \leq C/n$ for every $n \geq 1$ and every $p \in \mathbb{R}$.*

Proof. By Lemma 2.1, $C = \frac{1}{4}$ works. □

Proposition 2.6. *For $m \geq 2$ and $x, y \in \mathbb{R}^m$, $s^2(x+y) \leq 2s^2(x) + 2s^2(y)$.*

Proof. We have $s^2(x+y) = s^2(x) + s^2(y)$, and $s^2(x) + s^2(y) \leq 2s^2(x) + 2s^2(y)$ because both terms are nonnegative. □